Project No: A2100-261

Development of a Web-Page-Based Software Program for Technology-Enhanced Learning (TEL) of Combinational Logic Circuits

Project Plan / Strategy

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1. Motivation

Combinational logic circuits: multiplexers, decoders and encoders, are core topics in the Digital Electronics module (IE2104 / EE2004) at NTU EEE. They are currently taught through lectures and a fixed set of tutorial questions. Once a student has worked through the tutorial set, mastered the content and start doing past year exam papers, those striving to do well for the module will hit a brick wall of not having enough questions to practice on.

This is a problem because the skill being assessed is procedural rather than conceptual. Deriving the output function of a circuit built from a multiplexer and a decoder is a process: construct the truth table, trace which input is selected at each combination, evaluate the output, and compile the minterm or maxterm expression. Procedural skills improve through repetition and through immediate feedback when a step goes wrong. A limited amount of questions with answers at the back of the tutorial provides neither well.

A web-based tool that generates new questions on demand and checks answers automatically addresses both limitations directly. It provides a fast, easy way to effectively practice with an “unlimited” question pool as well as generation of answer as they further drill their concepts through repetitive practicing.

Taking a look and reviewing at past year senior’s approach onto this FYP topic on Technology-Enhanced Learning (TEL) projects:

- Tan, W. H. (2024) produced multiple-choice questions for 2-to-1 and 4-to-1 multiplexers and decoders.
- Poon Qi Chuan (2025) covered multiplexers in depth, with reference content, a configuration solver, and two quiz formats.
- Nurul Rushdina Binte Rosli (2025) addressed finite state machines.

Reviewing this prior work identifies three clear gaps:

- Encoders have not been covered by any project in the series.
- Decoders have been addressed only briefly, in a limited multiple-choice format.

- No existing tool generates questions in which more than one component appears within the same circuit. This is a recurring form in IE2104 examinations. For example, AY2024/25 Semester 1 Question 2(b) presents a circuit consisting of a decoder, a multiplexer and two logic gates, and asks for the minterms of the resulting function.

All three gaps are addressed by this project, with decoders and encoders as the primary focus.

A review of six IE2104 examination papers (AY2023/24 to AY2025/26) confirms some facts: Questions on combinational logic modules appear consistently as Question 2, worth approximately 25 marks, and fall into two forms:

1. Synthesis questions, which ask the student to implement a given function using a specified module and a limited set of gates

2. Analysis questions, which present a circuit and ask for the resulting minterm or maxterm expression.

2a. The analysis questions are where multiple components appear together. For instance, a decoder feeding a multiplexer, a priority encoder driving multiplexer select lines, a full adder driving select lines, or two multiplexers combined to form a full adder

2. Objectives

The objectives of this project are as follows, as of the project planning at this early stage (subjected to possible changes if improvements or simplification is needed, if the project time frame allows) :

1. Develop a web-page-based software program that runs in any modern browser without installation, hosted on a server and accessible to students on both desktop and mobile devices.
2. Cover the four selection and decoding modules taught in IE2104: decoders, encoders, multiplexers and demultiplexers, as two natural pairings. Decoders and encoders are the primary focus, as these were not covered in depth by previous projects in the series.
3. Automatically generate randomised practice questions in both forms used in IE2104 examinations: synthesis questions and analysis questions. This shall include questions in which more than one device appears within the same circuit, such as a decoder combined with a multiplexer, or a decoder combined with an encoder.
4. Display the internal truth table alongside each answer, including the intermediate signal values within the circuit, so that a student can identify which step of their working was incorrect rather than only being told the final answer.
5. Produce a complete final report and oral presentation documenting the design, algorithms and testing of the program.

3. Project Overview

3.1 Scope

- Analysis questions involving circuits with more than one device, following the forms found in IE2104 examinations. For example, a decoder feeding a multiplexer, or a decoder combined with an encoder.
- Decoders and encoders, covered in depth.
- Multiplexers and demultiplexers, completing the second pairing. A demultiplexer is a decoder with the data line driving the enable input, so it reuses the decoder implementation directly.
- Automatic answer verification, with worked solutions and truth tables.

3.2 Development Approach

The program will be built from scratch rather than by extending the previous project, so that all four components can be handled by one shared set of code rather than written separately.

It will be written in HTML, CSS and JavaScript and hosted on Firebase, following the previous projects in this series. Git and GitHub will be used for version control. A first page has already been deployed to confirm that the setup works.

JavaScript is a new language for me. I have prior experience in Python, and the early stages of the project are intended partly as time to become familiar with it.

The general idea is that each question is built from a circuit the program creates itself, using randomised values from a template. Because the program builds the circuit, it can work out the correct answer by going through every possible input combination and recording what the circuit produces. This also produces the truth table, which can then be shown to the student alongside the answer.

The exact structure of the code will be worked out during Stage 1, once I have studied the past-year papers and have a clearer picture of the question types involved.

3.3 Question Types

For each device, two question forms will be generated, following the pattern used in IE2104 examinations:

Synthesis Questions: given a Boolean function, implement it using a specified device and a limited set of logic gates.

Analysis Questions: given a circuit diagram of one or more devices, derive the resulting function in minterm or maxterm form.

Both forms will be generated for single devices and, once the individual devices are implemented, for circuits containing more than one device.

This follows the same two question types identified in the review of past IE2104 papers in Section 1, and mirrors the question format used in the previous multiplexer project.

4. Project Stages and Schedule

Period	Stage	Main Tasks	Official NTU FYP timeline
Sep – Oct 26	Stage 1	Learn web development, review the IE2104 past-year papers to identify the question types involved, and build the shared code that generates a circuit, evaluates it for every input combination, and produces the truth table.	**Project Plan/Strategy** — to Supervisor, Week 6 Sem 1 — **14 Sep 2026**
Oct – Dec 26	Stage 2	Add decoder and demultiplexer questions, circuit diagrams, answer checking and truth table display. Deploy a working version. Interim report and video presentation.	Interim Report (& Video Presentation) — to Supervisor and Examiner, Week 13 Sem 1, **10 Nov 2026**
Jan 27	Stage 3	Add binary encoders and priority encoders, reusing the code built in Stages 1 and 2.	—
Jan – Feb 27	Stage 4	Add multiplexers, reusing the existing code. Once all four devices are implemented, add questions in which more than one device appears within the same circuit.	—
Feb – Mar 27	Stage 5	Add brief reference content for each component, tidy up the interface, and test the generated questions and answers systematically.	**Draft Final Report** — to Supervisor, end of Week 10 Sem 2 — **25 Mar 2027**
Apr – May 27	Stage 6	Final fixes and report; demonstration and oral presentation.	**Final Report** — to Supervisor and Examiner, end of Week 12 Sem 2 — **9 Apr 2027** **Project Demonstration** — arranged with Examiner, Week 13 Sem 2 — **12–16 Apr 2027** **Oral Presentation** — after exams — **10–12 May 2027** **Full-Text Report** — to Library via NTU Digital Repository — **19 May 2027** **Final Report (revised after oral)** — to main Supervisor — **19 May 2027**

5. Summary

The deliverable of this project is a web-page-based program, accessible from any browser, that automatically generates practice questions from four of the combinational logic modules taught in IE2104: decoders, encoders, multiplexers and demultiplexers, and automatically checks the answers submitted by the user. Because the program builds each circuit itself from a randomised template, it can work out the correct answer and produce the internal truth table at the same time, so a student can see where their working diverged rather than only being told the final answer. Decoders and encoders are the main focus, as these were not covered in depth by previous projects, with multiplexers and demultiplexers completing the two pairings. Building all four on one shared set of code also allows questions to be generated in which more than one device appears within the same circuit, following the form used in IE2104 examination questions.